

MARY, Turkmenistan

WMO: 384573

Lat: 37.619N

Lon: 61.897E

Elev: 222

StdP: 98.69

Time Zone: 5.00 (E05)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-6.1	-14.8	1.1	-9.2	-10.9	1.5	0.0	9.2	12.5	8.1	9.4	2.2	90	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	41.8	20.2	40.1	19.6	38.9	19.2	21.6	37.3	21.0	36.4	20.4	35.5	4.8	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.8	26.5	15.2	11.1	26.6	14.2	10.4	26.6	63.8	37.3	61.3	36.1	59.1	36.1	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.4	-11.2	43.8	5.0	1.3	-14.8	44.7	-17.8	45.5	-20.6	46.2	-24.2	47.2		
(5)			-11.9	23.3	4.8	0.8	-15.4	23.8	-18.2	24.3	-20.9	24.7	-24.4	25.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	4.9	6.4	13.0	18.9	25.5	29.7	31.1	28.9	23.7	17.3	10.1	5.8	(6)
(7)		DBStd	10.32	5.59	5.77	5.07	4.37	3.01	2.48	2.52	2.81	3.27	4.54	5.30	4.70	(7)
(8)		HDD10.0	531	172	122	27	1	0	0	0	0	0	4	60	145	(8)
(9)		HDD18.3	1687	417	336	176	45	0	0	0	0	2	73	251	388	(9)
(10)		CDD10.0	3452	13	20	122	267	482	591	653	585	411	231	63	15	(10)
(11)		CDD18.3	1567	0	0	12	61	224	341	395	327	163	41	4	0	(11)
(12)		CDH23.3	22270	1	8	165	736	2792	5126	6192	4699	2000	497	51	3	(12)
(13)		CDH26.7	13324	0	2	49	316	1496	3248	4097	2928	999	174	16	0	(13)
(14)	Wind	WSAvg	3.3	3.3	3.5	3.6	3.7	3.5	3.6	3.3	2.9	2.7	2.7	3.1	(14)	
(15)	Precipitation	PrecAvg	151	18	23	28	23	15	4	1	1	7	16	14	(15)	
(16)		PrecMax	237	35	59	48	60	45	29	8	7	9	27	36	36	(16)
(17)		PrecMin	70	3	8	5	1	0	0	0	0	0	0	2	1	(17)
(18)		PrecStd	45	8	15	11	16	12	6	2	2	2	6	10	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	24.1	31.8	36.2	39.2	42.1	44.1	42.2	38.8	34.9	29.0	22.9	(19)
(20)			MCWB	10.8	13.6	16.0	18.2	19.4	20.3	20.7	20.3	18.5	17.5	15.5	11.0	(20)
(21)		2%	DB	18.9	20.0	27.2	33.0	37.1	40.2	42.2	40.1	36.2	30.9	24.0	18.9	(21)
(22)			MCWB	10.0	11.0	14.7	17.6	18.7	19.6	20.4	19.4	17.7	16.1	13.0	10.7	(22)
(23)		5%	DB	15.2	17.1	24.8	30.1	35.8	39.1	40.9	38.9	34.1	28.2	21.0	15.8	(23)
(24)			MCWB	9.0	10.6	14.0	16.1	18.2	19.1	19.9	19.0	17.4	15.3	12.2	9.3	(24)
(25)		10%	DB	12.2	14.8	21.9	27.8	33.9	37.9	39.1	37.2	32.1	26.0	18.2	13.0	(25)
(26)			MCWB	7.4	9.4	13.0	15.8	17.8	18.9	19.5	18.6	17.0	14.4	10.8	8.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	14.0	17.5	20.2	21.2	22.2	22.9	22.1	20.2	18.6	16.4	12.4	(27)
(28)			MCDWB	19.2	22.2	27.8	31.8	34.4	37.8	38.0	37.3	33.8	31.7	26.5	20.0	(28)
(29)		2%	WB	10.7	12.3	15.7	18.6	20.2	21.2	21.8	21.0	19.2	17.2	14.4	10.9	(29)
(30)			MCDWB	17.6	18.6	24.7	29.6	33.4	36.9	38.0	36.1	32.8	28.4	21.7	17.3	(30)
(31)		5%	WB	9.3	10.7	14.5	17.4	19.4	20.5	21.2	20.2	18.3	16.1	12.8	9.7	(31)
(32)			MCDWB	15.2	15.7	23.1	27.2	32.1	35.8	37.2	35.5	31.9	26.3	19.0	15.1	(32)
(33)		10%	WB	7.6	9.8	13.5	16.3	18.7	19.8	20.5	19.4	17.5	15.1	11.5	8.4	(33)
(34)			MCDWB	11.9	14.2	21.0	25.2	30.5	34.8	36.3	34.8	30.5	25.0	17.5	12.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	10.0	11.5	12.5	13.7	14.6	15.3	15.2	13.9	11.6	9.6	(35)	
(36)			MCDBR	14.7	14.4	16.5	17.7	16.4	16.4	16.9	16.8	17.8	17.2	15.6	14.5	(36)
(37)			MCWBR	7.9	7.9	6.8	6.9	5.5	5.2	5.5	5.8	6.7	7.6	7.9	8.0	(37)
(38)		5% WB	MCDWB	13.9	12.6	15.1	14.9	14.4	14.8	15.5	15.3	15.8	15.4	14.1	13.3	(38)
(39)			MCWBR	7.9	7.3	6.9	6.7	5.2	5.1	5.4	5.6	6.5	7.1	7.5	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.391	0.437	0.508	0.540	0.542	0.548	0.532	0.487	0.459	0.488	0.448	0.398	(40)	
(41)		taud	2.086	1.971	1.827	1.799	1.817	1.796	1.837	1.925	1.963	1.874	1.953	2.061	(41)	
(42)		Ebn at Noon	754	766	747	754	761	755	763	789	784	700	676	709	(42)	
(43)		Edh at Noon	120	153	195	212	212	216	206	184	167	166	137	116	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.25	3.21	4.35	5.74	7.09	8.02	7.90	7.30	6.03	4.24	2.75	2.07	(44)	
(45)		RadStd	0.19	0.33	0.34	0.49	0.26	0.24	0.17	0.18	0.15	0.24	0.24	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	-0.58	-0.46	N/A	N/A	N/A		

Nomenclature: See separate page